

CASE STUDY

Groceries for Good

A Sustainability-Focused Grocery Shopping App

ROLE	ORGANISATION	TEAM	TIMELINE
Project Manager	Visa Hackathon	EternalGarden	June 2024

Challenge: Design an app that allows grocery store customers to understand their environmental impact associated with their purchases, learn how they can reduce their impact, and be rewarded through a gamified experience.

01 — Project Overview

Groceries for Good is a consumer-facing mobile application built during the Visa Hackathon. The product uses anonymised payment and transaction data to help grocery shoppers understand the environmental footprint of their purchases, discover sustainable alternatives, and earn rewards for making eco-friendly choices.

My Role as Project Manager

As Project Manager for Team EternalGarden, I was responsible for:

- Defining the project scope, sprint structure, and delivery milestones across the hackathon timeline
- Coordinating research, design, and development workstreams to stay on schedule
- Facilitating brainstorming sessions, user research planning, and synthesis workshops
- Ensuring the team’s output aligned with the judging criteria: Feasibility, Innovation, Desirability, and Scalability
- Owning stakeholder communication and final presentation structure

Judging Criteria

Criterion	Description
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Feasibility	Can the project be completed given time, resources, and technology?
Innovation	Is this a unique approach or new type of solution?
Desirability	Is the solution attractive to users and does it meet an unmet need?
Scalability	Can the solution scale over time and can success be measured?

02 — Secondary Research

Before talking to users, the team conducted desk research to ground our understanding of the problem space. I coordinated this phase to ensure we had a shared factual foundation before primary research began.

Food Supply Chain & Environmental Impact

Production & Distribution

- Crops grown on farms, processed through facilities, transported via trucks, trains, ships, and planes to retailers
- Agriculture accounts for a significant portion of global freshwater usage, contributing to water scarcity
- Intensive farming leads to deforestation, habitat loss, and soil degradation

Key Environmental Impacts Identified

- **Transportation:** Greenhouse Gas Emissions
 - Long-distance food transportation from fossil fuel vehicles
 - Energy-intensive processing and refrigeration at facilities
- **Food & Packaging:** Waste Generation
 - Food loss occurs at every stage from farm to fork
 - Excessive plastic, cardboard, and Styrofoam packaging pollutes landfill
- **Agricultural:** Chemical Pollution
 - Pesticides and fertilisers contaminate soil, waterways, and ecosystems
 - Processing chemicals present environmental risks if unmanaged
- **Loss of Biodiversity:** Monoculture farming and habitat destruction reduce ecosystem resilience
- **Climate Change:** Disrupts agriculture through temperature shifts, altered precipitation, and extreme weather events
- **Social & Economic Impacts:** Unfair labour conditions in food supply chains create economic disparities

Key insight from research: Grocery shopping sits at the intersection of all these environmental harms. Individual consumer choices, multiplied at scale, represent a significant lever for reducing environmental impact — if consumers have the right information and incentives.

03 — User Research

I led the planning and facilitation of four in-depth user interviews to understand real motivations, behaviours, and pain points around sustainable grocery shopping. Each interviewee represented a distinct user archetype.

Interview Methodology

Each interview explored five key areas:

- Understanding motivations — what drives sustainable shopping behaviours
- Prior experiences with sustainable product discovery
- The role of sustainability relative to price and convenience
- Challenges and barriers encountered
- Goals, aspirations, and desired improvements to the grocery experience

04 — Analysis & Synthesis: User Personas

Following interviews, I facilitated a synthesis session to build four representative personas. These served as the team’s decision-making compass throughout design and development.

Persona 1: Alex — The Sustainability Coordinator

Alex works at a local sustainability organisation and is deeply committed to environmental advocacy. He approaches grocery shopping as an extension of his professional values.

Alex — Sustainability Coordinator (28)		
Goals • Advocate sustainable practices in community • Inspire eco-friendly choices	Challenges • Navigating overwhelming information for sustainable products	Key Insights • Believes in community engagement for meaningful change

• Reduce environmental footprint in grocery shopping • Empower individuals for positive environmental change	• Balancing affordability with eco-friendly options • Limited availability of eco-friendly products	• Uses technology for broader outreach and community mobilisation • Envisions streamlined shopping with clear labelling and personalised recommendations
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Persona 2: Maya — The Committed Activist

Maya, 27, is an environmental activist for whom sustainability is a top priority across every aspect of life. She actively seeks to lead by example and support environmentally aligned companies.

Maya — Environmental Activist (27)		
Goals • Lead by example in sustainable living • Prioritise sustainability in grocery shopping • Advocate for accessible, eco-conscious everyday choices	Challenges • Difficulty finding aligned products in traditional stores • Balancing affordability with higher eco-friendly prices • Frustration with lack of information and options for sustainable shopping	Key Insights • Driven by values and desire to lead by example • Believes grocery stores can promote sustainability through education and incentives • Envisions convenient, accessible shopping with clear labelling and personalised recommendations

Persona 3: David — The Responsible Teacher

David, 45, is a teacher who prefers in-store grocery shopping. He wants to set a positive sustainability example for his students and understands the environmental impact of his choices, though faces practical constraints.

David — Teacher & Role Model (45)		
Goals • Reduce environmental footprint through sustainable grocery shopping • Set example for students; educate on sustainability	Challenges • Discerning greenwashing from genuine sustainability • Balancing sustainability with budget limitations	Key Insights • Driven by commitment to sustainability as a teacher and role model • Desires clear labelling and incentives for eco-friendly choices

• Ensure products benefit both family and planet	• Addressing frustration with excessive packaging	• Envisions accessible shopping with personalised recommendations
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Persona 4: Emily — The Overwhelmed Pragmatist

Emily, 32, is a marketing manager who shops both in-store and online. She cares about sustainability but is overwhelmed by the lack of accessible, clear information, making it hard to consistently prioritise eco-friendly choices.

Emily — Marketing Manager (32)		
Goals • Reduce environmental footprint through shopping habits • Make choices aligned with personal values of environmental stewardship • Contribute to a healthier planet for future generations	Challenges • Difficulty navigating overwhelming information on sustainable options • Struggle to find affordable sustainable alternatives • Frustration with excessive plastic packaging in mainstream products	Key Insights • Prioritises sustainability in grocery shopping for future generations • Lack of accessible information hinders sustainability efforts • Values transparency from brands on environmental practices • Believes grocery stores can promote sustainability through education and incentives

05 — Common Themes

After synthesising insights across all four personas and interviews, five consistent themes emerged. These themes directly informed our product strategy.

01	Motivation All users share a desire to align their actions with their values and make a positive impact on the environment. This is the core driver — not just awareness, but a genuine desire to act.
02	Awareness Gap There is a clear call for better education and awareness from both consumers and grocery stores about the environmental impact of products.
03	Shopping Experience

04	Everyone desires a simpler, more accessible shopping experience where sustainable options are clearly labelled and readily available.
	Challenges A universal consensus on four barriers: information overload, affordability, limited availability of eco-friendly products, and time constraints.
	Community Collaboration and community engagement are seen as crucial in promoting sustainability and making a collective impact.

06 — Competitor Analysis

I assigned team members to research eight existing products in the sustainable shopping and carbon tracking space. Understanding the competitive landscape allowed us to identify white space and sharpen our unique value proposition.

Competitor	Strengths	Weaknesses
Giki	Mobile barcode scanner; personalised sustainability recommendations; gamified rewards and badges	Limited by scope and availability of product data
Yuka	Scans barcodes for health and sustainability ratings; strong user engagement and comprehensive database	Primarily focused on health — less emphasis on carbon footprint
Doconomy	Integrates transaction data for carbon footprint measurement; API (Aland Index) already works with Mastercard	No guidance on daily purchasing patterns; database not specific to product level
Oreco	Tracks lifestyle impact on climate; community features; badges and challenges; personalised insights	Targeted to environmentally conscious users; carbon calculation accuracy limited by user data
GoodGuide	Comprehensive product ratings (environmental, social, health); barcode scanning; established platform	Lack of personalisation; limited coverage of niche products; subjectivity in ratings
EcoCart	Browser extension with sustainability ratings per product; robust data from scientific studies and NGOs	Online shopping only; no in-store use; no education or broader sustainability tips
Karma	In-store barcode scanning; loyalty card integration; comprehensive research on product impacts; partnerships with grocery chains	UK/EU only; lack of transparency in ratings; no social impact data

Abillion	Community-driven reviews; covers carbon, packaging, and health; brand-level assessment; barcode scanning	Inconsistent and unverified data; bias/lack of transparency; not personalised
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Our gap: No existing competitor combines receipt/transaction data (Visa’s unique asset), in-store and online coverage, vendor-level Scope 1–3 emissions data, AND a gamified rewards programme — all in a single consumer-facing app.

07 — Journey Mapping

The team created a journey map for David (Persona 3) — chosen as the most representative ‘middle-ground’ user. The map traced his experience across six stages and surfaced the most critical pain points and opportunities.

Stage	Discovery	Onboarding	Exploring	Shopping	Checkout	Tracking
Pain Points	Information overload when searching	Unclear sustainability claims on onboarding	Greenwashing hard to identify	Budget constraints vs. eco options	Excessive packaging at checkout	No feedback loop on impact over time
Opportunities	Clear app introduction with value prop	Personalised onboarding quiz	AI-powered sustainability ratings	Affordable eco-alternative suggestions	Carbon emission display at checkout	Monthly impact dashboard & rewards

08 — Ideation

I facilitated two structured ideation exercises with the team: a Brainstorming Session (10 min) and a Rolestorming exercise (10 min), using a digital sticky-note board to capture all ideas without judgement.

Top Ideas Generated

- Connect card to app: flag if recent purchases don’t meet sustainability score; redirect credits to sustainable projects
- Scan receipt to surface sustainability impact of all products in the basket; show carbon footprint with Ecológiq data
- Reward system: sustainability points grow like a tree; redeem for eco-friendly incentives
- Loyalty programme integration with Visa to convert points for sustainable choices

- In-app product database for searching sustainable alternatives and building shopping lists
- Show shelf-life and waste-reduction tips to minimise food waste
- Community leaderboards: friends and family competing on sustainability scores
- Monthly AI-generated sustainability recommendations for reducing carbon footprint
- Vendor portal: enable vendors to submit Scope 1, 2, 3 emissions data to enrich product ratings
- Daily donation/investment to selected green projects based on shopping behaviour
- Partner with local farmers and sustainable brands; surface them at top of search results
- Impact reports and local sustainability challenges to drive community engagement

09 — MVP Feature Prioritisation

From ideation, I led the team through a feature prioritisation exercise to define three MVPs. Each was scoped with clear objectives and development steps to ensure delivery within the hackathon timeline.

MVP 1: Receipt Submission App for Carbon Emissions & Sustainable Alternatives

Objective: Allow customers to submit grocery receipts, understand their carbon emissions, and receive sustainable alternative suggestions.

1. Receipt Submission Feature

- OCR integration or AI: scan and process receipts
- Data extraction: itemise product names, quantities, and prices

2. Carbon Emission Calculation

- Database integration: associate products with carbon footprint data
- Emission calculation algorithm: calculate total footprint from receipt data

3. Sustainable Alternatives Suggestions

- Alternative products database: maintain a catalogue of sustainable alternatives
- Recommendation engine: suggest alternatives based on purchase history

4. User Interface & Experience

- Dashboard: display carbon emission data clearly
- Feedback loop: allow users to rate suggestions and improve recommendations

MVP 2: Gamification Around a Reward Programme

Objective: Implement a gamified reward programme to encourage sustainable shopping behaviours.

1. Reward System

- Points system: earn points for sustainable purchases, receipt submissions, and eco-friendly actions
- Tiered rewards: increasing benefits and incentives as users level up

2. Gamification Elements

- Challenges and quests: weekly or monthly sustainability targets (e.g. reduce carbon by 10%)
- Badges and achievements: awarded for completing sustainability milestones

3. Social Features

- Leaderboards: foster friendly competition among users
- Community challenges: group participation with collective impact tracking

4. Redeemable Rewards

- Discounts, free products, and exclusive offers for points
- Access to exclusive sustainability content, events, or workshops

MVP 3: Vendor Capability for Scope 1, 2, & 3 Data Submission

Objective: Enable vendors to upload their Scope 1, 2, and 3 emissions data to enhance product-level accuracy and consumer trust.

1. Vendor Portal Development

- Secure login portal for vendors to access and manage their data
- Easy-to-use data upload interface with clear Scope definitions

2. Emissions Data Integration

- Data validation checks to ensure accuracy and completeness
- Guidelines for Scope 1, 2, and 3 data submission provided to vendors

3. Data Utilisation

- Enhanced recommendations: use vendor data for more accurate product-specific sustainability ratings
- Vendor insights: show vendors their sustainability performance and improvement suggestions

4. Collaboration & Feedback

- Vendor feedback loop to share experiences and improve the system
- Regular updates to vendors on how their data is used and its impact on consumer choices

10 — Product Development Plan

As Project Manager, I defined and managed the sprint structure for the hackathon. All sprints were contained within June 2024.

Sprint	Dates	Focus
Sprint 1	Jun 3–6 & Jun 6–11	Data Gathering, User Research → Analysing & Design
Sprint 2	Jun 6–11	Continued analysis and design iteration
Sprint 3	Jun 10–11 & Jun 11	User Testing, Dev Contact & Dev Handoff
Sprint 4	Jun 11–19	Development, User Testing (Jun 18–19), Presentation (Jun 19–20)

11 — Reflections & Key Learnings

What Worked Well

- Structured research phases: front-loading user interviews and secondary research before ideation ensured all product decisions were evidence-led
- Persona-driven design: the four personas served as consistent reference points that kept the team aligned on user needs throughout
- Competitive landscape clarity: mapping eight competitors early revealed genuine white space for our differentiated value proposition
- Sprint structure: breaking the hackathon into focused sprints helped manage scope and maintain momentum under time pressure

Challenges Overcome

- Information overload in the problem space required me to facilitate tight synthesis sessions to extract signal from noise
- Balancing breadth of features against hackathon time constraints meant making tough prioritisation calls in the MVP phase

- Aligning a multidisciplinary team across research, design, and development required clear communication and shared documentation

Key Takeaways as PM

- User research is non-negotiable, even in time-pressured environments — it de-risks assumptions and generates sharper solutions
- A clear mission statement ('Revolutionise the use of payments data to drive sustainable outcomes') serves as a powerful north star for the whole team
- Competitor analysis is most valuable when synthesised into a clear gap statement, not just a feature list comparison
- Gamification works best when tied to genuine user values — not as a superficial overlay but as a reflection of what users already care about

Groceries for Good demonstrates that payment data, used thoughtfully, can be a powerful tool for environmental good — turning every grocery trip into an opportunity to learn, improve, and be rewarded for making a difference.